

# eto Optimization -- Product Design Trade-

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Mode: Full Pipeline

# 1. Problem Formulation

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**Problem Type:** Find  
**Domain:** Multi-Objective Optimization

## Universe

- Knee Point:  $\{\text{cost, inv\_durability, thickness, alloy\_grade, coating\_level, frontier\_index}\}$
- Utopia:  $\{\text{cost, inv\_durability}\}$
- Nadir:  $\{\text{cost, inv\_durability}\}$

## Variables

| Symbol | Meaning           | Type / Range |
|--------|-------------------|--------------|
| x      | decision variable | $\mathbb{R}$ |

## Structure

A manufacturing company designs a product with two competing objectives: minimize cost and maximize durability (equivalently, minimize inverse durability). There are 3 design variables -- material thickness, alloy grade, and coating level -- each with physical bounds. The goal is to find the Pareto frontier (the set of non-dominated designs), identify the knee point (best compromise), and quantify the trade-off rates along the frontier.

**Design Variables:**

- Material thickness: [1.0, 10.0] mm
- Alloy grade: [1.0, 5.0] (index)
- Coating level: [0.5, 3.0] (index)

**Objectives:**

- Minimize cost:  $f_1(x) = 2 \cdot \text{thickness} + 5 \cdot \text{alloy}^2 + 3 \cdot \text{coating}$
- Minimize inverse durability:  $f_2(x) = -\ln(\text{thickness}) - 2 \cdot \text{alloy} - 1.5 \cdot \text{coating}^{0.8}$

## Real-World Mapping

| Real-World Concept | Mathematical Object |
|--------------------|---------------------|
| problem input      | instance parameters |

## Constraints

- (1) Pareto optimality\*\*: A solution is Pareto-optimal if no other feasible solution improves one or more objectives without worsening at least one other.
- (2) Epsilon-constraint method\*\*: Convert a multi-objective problem into a sequence of single-objective problems by optimizing one objective while constraining the others.
- (3) Knee point\*\*: The point on the Pareto front where the trade-off rate changes most sharply, often the best practical compromise.
- (4) Utopia point\*\*: The ideal (usually infeasible) point formed by the best value of each objective function.
- (5) Nadir point\*\*: The worst objective values among Pareto-optimal solutions, defining the "box" bounding the Pareto front.
- (6) Marginal rate of substitution\*\*: How much of one objective must be sacrificed to gain a unit increase in another.
- (7) Non-dominance\*\*: The defining property of Pareto-optimal solutions: no other solution is at least as good in every objective.

## Approach

**Suggested approach** Epsilon-constraint + Pareto filter (scipy L-BFGS-B)

**Complexity class:**

**Available tools:** Epsilon-constraint + Pareto filter

## 2. Solution

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|                   |                                                     |
|-------------------|-----------------------------------------------------|
| <b>Answer:</b>    | Solution computed                                   |
| <b>Optimal:</b>   | No / Unknown                                        |
| <b>Feasible:</b>  | No                                                  |
| <b>Algorithm:</b> | Epsilon-constraint + Pareto filter (scipy L-BFGS-B) |
| <b>Time:</b>      | 2.3154s                                             |

### Solution Details

Method: 1. Individual optima: Minimize each objective independently to establish the feasible objective range

2. Epsilon-constraint sweep: Fix one objective as a constraint (durability  $\geq$  epsilon) and optimize the other (cost), sweeping epsilon across the feasible range

3. Weighted-sum comparison: Solve scalarized problems  $\min w_1*f_1 + w_2*f_2$  for a grid of weights

4. Pareto filter: Remove dominated points -- a point is dominated if another point is at least as good in all objectives and strictly better in at least one

5. Knee point: Find the point on the Pareto front with maximum curvature (largest angle formed by its neighbors), indicating the best marginal trade-off

6. Utopia/nadir: The utopia point is the (infeasible) vector of individually optimal objectives; the nadir point is the worst objective value attained by any Pareto-optimal solution in each dimension

7. Trade-off rates: Compute the marginal rate of substitution ( $\Delta f_2 / \Delta f_1$ ) between consecutive Pareto points

### Verification

No independent verification checks recorded.

## 3. Interpretation

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### The Question

A manufacturing company designs a product with two competing objectives: minimize cost and maximize durability (equivalently, minimize inverse durability).

### The Answer

**A feasible solution was found.**

### What This Means

- 50 candidate solutions generated via epsilon-constraint sweeps
  - Pareto front filtered to ~20-40 non-dominated points
  - Weighted-sum solutions for comparison
  - Knee point identified at the point of maximum curvature
  - Utopia point (infeasible ideal) and nadir point (worst of the bests)
  - Trade-off rates (marginal rate of substitution) along the frontier
  - Full independent verification (non-dominance, bounds, objective recomputation)
  - Solution saved to solution.json
1. Individual optima: Minimize each objective independently to establish the feasible objective range
  2. Epsilon-constraint sweep: Fix one objective as a constraint (durability  $\geq$  epsilon) and optimize the other (cost), sweeping epsilon across the feasible range
  3. Weighted-sum comparison: Solve scalarized problems  $\min w_1*f_1 + w_2*f_2$  for a grid of weights
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## Recommendations

1. Review the model assumptions to ensure real-world fidelity.

## Limitations

- Model assumes deterministic parameters; real-world variability not captured.

- Solution